Rewarding Learning

ADVANCED SUBSIDIARY (AS)
General Certificate of Education
2026

Centre Number

Candidate Number

Digital Technology

Assessment Unit AS 2
assessing
Fundamentals of
Digital Technology

SDT21

[SDT21]
TUESDAY 26 MAY, MORNING

TIME

1 hour 30 minutes.

INSTRUCTIONS TO CANDIDATES

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
You must answer the questions in the spaces provided.
Do not write outside the boxed area on each page or on blank pages.
You are not permitted to use any calculating aid in this paper.
Complete in black ink only. **Do not write with a gel pen.**
Answer **all five** questions.

INFORMATION FOR CANDIDATES

The total mark for this paper is 100.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each question or part question.
Quality of written communication will be assessed in Questions **2(d)**, **3(d)** and **5(e)**.

15025

1 **(a)** Write down the range of decimal values that can be represented using 4 bits.

Smallest decimal value

_____

Largest decimal value

_____ [2]

(b) How many bits are in 2 kilobytes?

You must show all your working.

_____ bits [2]

(c) What is the least number of bits required to represent the decimal number **28** in binary.

You must show all your working.

_____ bits [2]

(d) Convert the decimal number **68** into an 8-bit binary number.

You must show all your working.

_____ [2]

(e) Convert the decimal number **–22** into a two's complement binary number using 8 bits.

You must show all your working.

_____ [3]

(f) Convert the two's complement binary number $\mathbf{10000110_2}$ to decimal.

You must show all your working.

_____ [3]

[Turn over

2 **(a)** The results of a Digital Technology test are recorded in a table. An extract from the table is shown below.

Student Number	Mark (%)	Extra Tuition Needed?
S101	65	N
S102	45	Y

The pass mark for the test was 50%.

(i) By referring to the shaded cell, explain the following terms.

Data

Information

[4]

(ii) By referring to the test result for student **S102**, explain what is meant by knowledge.

[3]

[Turn over

(b) The following table shows the hours worked by supermarket staff in a week and their net pay.

StaffID	Hours Worked	Net Pay (£)
200	22	220
201	25	250
202	15	150
203	12	120

When the data from this table is entered into the payroll system, two totals are calculated and stored along with the data. These totals are **4** and **740**.

By referring to the two totals, explain the difference between a control total and a hash total.

[4]

(c) One of the product codes in a stock control system is **57249**.

Calculate the modulus 11 check digit which should be added to this product code.

You must show all your working.

Check digit _____

[5]

[Turn over

(d) Data verification can be used when data is entered into a computer system.

Discuss the effectiveness of data verification in ensuring that the data entered is correct. You should refer to two different methods of data verification.

Quality of written communication will be assessed in this question.

[6]

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(Questions continue overleaf)

[Turn over

3 **(a)** A student plans to buy a computer which will be used mostly for gaming.

The following table shows part of the specification of two popular PCs.

Computer A	**Computer B**
RAM 8 GB DDR4 (2933 MHz) **Processor** Quad-core 12 MB cache	**RAM** 16 GB DDR4 (3600 MHz) **Processor** 12-core 25 MB cache

(i) By referring to the table, describe two reasons why the student should select **Computer B**.

1. _____

_____

_____

_____

2. _____

_____

_____

_____ [4]

(ii) Explain why a computer requires RAM as well as secondary storage.

RAM

Secondary storage

[6]

(iii) Identify three reasons why flash storage is suitable for use as secondary storage.

1.

2.

3.

[3]

[Turn over

(b) Describe the main features of each of the following processor components.

Clock

Cache memory

[4]

(c) The following table describes how some components are used during the fetch-execute cycle.

Complete the table by inserting the name of the relevant components.

Description	Component
Stores data as it is fetched from memory	
Transfers commands from the CPU to other components	
Is incremented to indicate the next instruction to be executed	

[3]

(d) The computer system used by a large organisation supports both multiprocessing and multitasking.

Distinguish clearly between multiprocessing and multitasking by referring to how each type of operating system could be used in a computer system.

Quality of written communication will be assessed in this question.

[6]

4 **(a)** Describe the purpose of an operating system.

[2]

(b) Describe the main features of each of the following utility programs.

File backup

Archive software

[4]

(c) Explain what is meant by a forms dialogue user interface.

[2]

(d) Airports use real-time processing systems for air traffic control.

Explain why a real-time processing system is required for air traffic control rather than a batch processing system.

[4]

(e) An organisation requires a new payroll system. They must decide between a custom-built solution or an off-the-shelf solution.

(i) State two advantages of purchasing a custom-built solution.

1.

2.

[2]

(ii) State two advantages of purchasing an off-the-shelf solution.

1.

2.

[2]

[Turn over

5 **(a)** Consider the domain name **www.ccea.org.uk**.

The domain name contains four parts. The table below identifies each of these parts.

Complete the table using the letters **A** to **D**. Each letter should only be used once.

A **Second level domain**
B **Top level domain**
C **Domain name**
D **The country level domain**

Part of domain name	Letter
www.ccea.org.uk	
.org	
ccea.org	
uk	

[4]

(b) Each department in a school has its own web page.

For example, the Digital Technology department uses the web page

http://www.clarendonhigh.ac.uk/digitaltechnology

By referring to this example, explain the difference between a domain name and a URL.

_____

_____

_____

_____

_____

_____ [4]

15025

(c) Consider the following segment of HTML code. Each line is numbered for reference.

```
1  <h2>Topics</h2>
2  <ul>
3     <li>Data</li>
4     <li>Information</li>
5     <li>Multimedia</li>
6  </ul>
```

The following changes are required.

1. The word Topics should be larger
2. The list of topics should be shown as a numbered list

Show the modifications which should be made to this code to implement each of these changes.

Change 1

_____

_____

Change 2

_____

_____ [4]

[Turn over

(d) Students and teachers can access their school's intranet.

Describe how the intranet could be used by teachers for each of the following.

Learning resources

Communication

[4]

(e) A school has introduced an online shop for pupils.

Pupils use an online form to

- Enter their username and password
- Select the items they wish to purchase
- Pay for purchases

The form provides pupils with feedback if they have not completed all sections of the form or when an item is not in stock.

Pupils receive an email to confirm their order has been accepted.

Using examples from this online application, distinguish clearly between client-side processing and server-side processing.

Quality of written communication will be assessed in this question.

[6]

THIS IS THE END OF THE QUESTION PAPER

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Question Number	**Marks**
1	
2	
3	
4	
5	
Total Marks	

Examiner Number

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